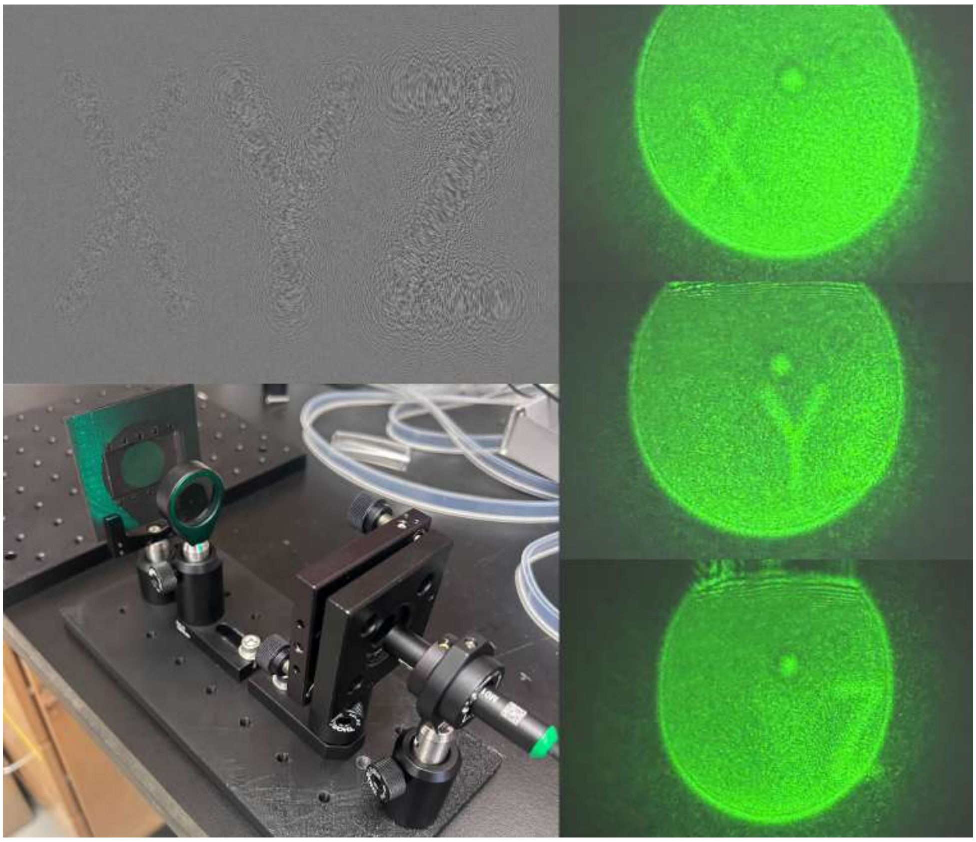

LabEscape Multi-depth Holography

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Final Report **November 2025**

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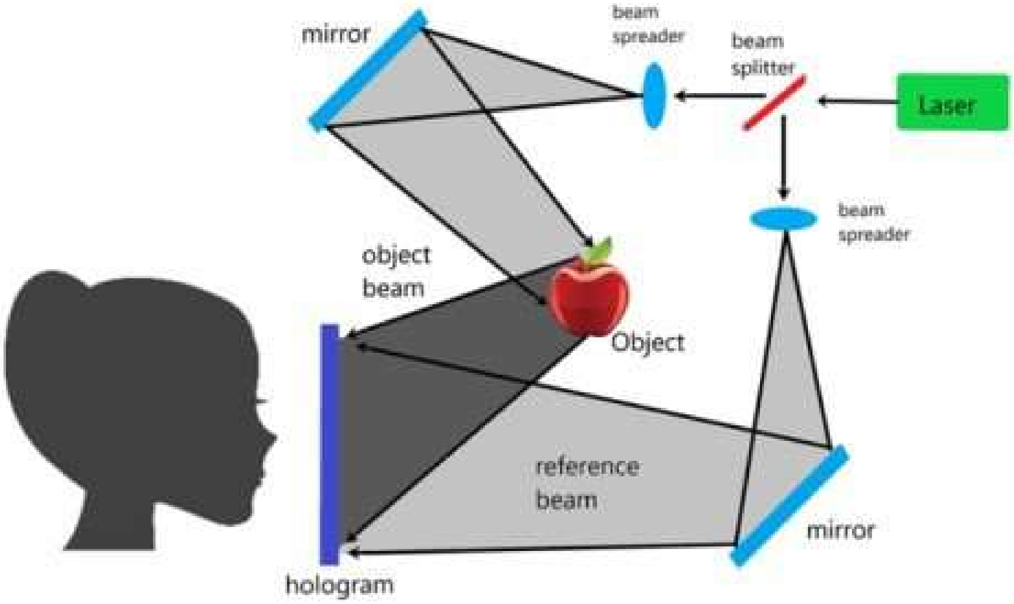
1 Introduction

This is a report of a year-long project to create multi-depth Computer-Generated Holograms (CGH) for *LabEscape*, currently used as of October 2025. It details both multi-depth and single-depth holography, both of which used many kinds of software and hardware to create the holograms, mounts, and components. We detail the steps toward the final setup, including how to create holograms, develop the film, and make possible improvements for future development.

1.1 Theory and Past Work

Holography First developed almost a century ago, holography operates through "storing" interference patterns. As two coherent beams mix, their electric fields add like $|E_1+E_2|^2 = |E_1|^2 + |E_2|^2 + 2Re\{E_1E_2^*\}$. The last term is the "interference term." Materials such as holographic plates have an emulsion layer to store the information contained in this interference term physically. This includes a "reference beam" and "object beam," as shown in Figure 1:

Figure 1: **Recording physical holograms.**



Instead of using physical holography, we opted to use computer-generated holography. We used code based on Python and MATLAB, each with different methods of generating holograms, to create custom patterns and images. For more information on physical holography, I suggest reading tutorials on sites such as *Integraf*, and watching the near-complete explanation of holography by 3Blue1Brown on *YouTube*.

Past Work Multiple undergraduate and graduate students used film and holographic materials to perform experiments in the past, however, not much of their work was documented. A lot of time was dedicated to recovering this information, with mixed results. In particular, Julio Barreiro and Rachel Hillmer used holography to create and store different quantum states. They used black and white Kodak film and a film printer to create patterns that act as amplitude gratings. Then, they expose the film pattern onto a holographic plate, creating a phase grating.

Despite many months of troubleshooting, using holographic plates yielded poor image quality and efficiency. We opted to use the Kodak TMAX 100 black and white film instead, creating amplitude gratings.

2 Necessary Hardware

We detail the devices used for this project, alongside instructions of how to use each:

2.1 Spatial Light Modulator: Holoeye PLUTO

Figure 2: **The PLUTO SLM.**

We own a PLUTO SLM from 2008. It is a reflective phase-only SLM, meaning it will change the phase of the light incident on its screen according to the pattern given to it, with each color value of the image (0-255) corresponding to a phase shift, thereby interfering portions of the beam to create images in the far field.

Communication and Installing Software The SLM connects to a computer with two ports: an RJ11 and DVI port. The RJ11 port is used for changing firmware, and the DVI port is used to connect a display. We used an RJ11 to RS232, then a Serial-USB cable:

Figure 3: **The necessary cable for the SLM: RS-232 to RJ11, into a USB cable.**

After obtaining this cable, we downloaded software from the Holoeye website with a registered account. You can register on the website using the SLM serial number by email. This allows you to access configuration files, manuals, and software. It also has code to display specific images using Python, MATLAB, or LabView.

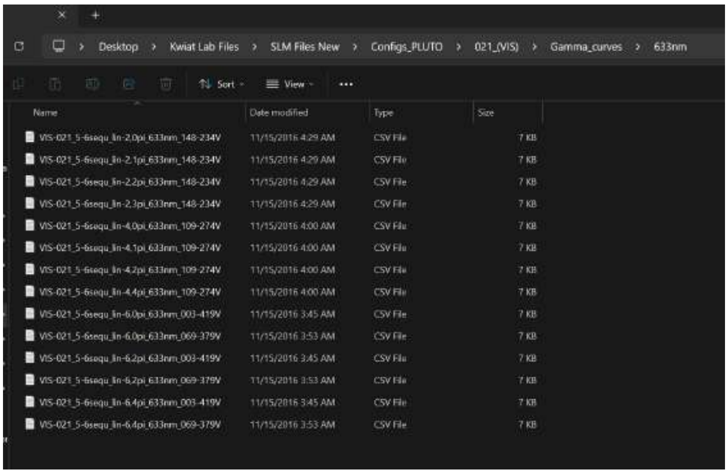
SLM Pattern Generator The Pattern Generator software controls the display of the SLM. It contains features such as displaying image files, creating holograms through the software, and "Elementary Optical Functions" like lenses and apertures. We describe its use later.

SLM Configuration Manager Configuration Manager should be used as little as possible. It can calibrate the SLM for specific wavelengths. Calibration is described in the PLUTO manual and below:

Calibrating the SLM *If you wish to calibrate the SLM, PLEASE READ before attempting. Only calibrate if you deem absolutely necessary.* The SLM has specific calibration tables corresponding to different wavelengths used. The software assigns a specific voltage value for each 0-255 grayscale color, where pure black (0) corresponds to no phase shift and a maximum phase shift at 255; however, the realized phase shifts will be much different for red light than for blue light. You can change the SLM configurations through the "Pluto_Configuration_Manager" application in the folder downloaded from the Holoeye website.

After downloading the application, connect the SLM to the computer using the serial cable and HDMI-DVI cable. Then, choose your configuration. The gamma curves are detailed in the "Configs_PLUTO" folder. Navigate to a folder according to your wavelength (VIS = visible, NIR = near infrared, so on). For example, this is a list of files for 633nm configurations:

Figure 4: **Locating gamma curves.**



We chose a curve with "2pi" since we only required a 2π phase shift. For exact reference on the best curve or SLM settings, send an email to Holoeye. They were very receptive to any issues we faced and were a great help through the process.

Once you select your gamma curve and connect the SLM to the computer using Configuration Manager, navigate to File -> Load Gamma Curve. Then, select the .csv file and wait. We consistently got an "NVRAM failure" issue, and when we checked the current gamma curve (Edit -> Gamma Function), the values were randomized. We found that it took more than 10 attempts to successfully load the gamma curve; however, we did not encounter issues once it worked. I recommend only calibrating the SLM if you get poor results with the current gamma curve and must change it.

2.2 The Film Printer: Lasergraphics Mark III

Figure 5: **The Lasergraphics Mark III.**

The film printer is the basis of the entire project: it is how we convert digital images to physical patterns on film. *Here* is the link to the Lasergraphics manual, and *here* are some more specs and information regarding the printer. We detail specific settings and how to operate it later.

2.3 Film: Kodak TMAX 100

In the past, people used Kodak TMAX 100 for the film printer, so we did as well. *Here* are the specifications for it, although we never had to consider any specs or had issues with the film, as it is standard 35mm black and white film.

2.4 Developing Materials

To develop the film, you need specific chemicals, containers, and tools. I stored these close to the sink in room 388–the containers above the sink and chemicals below. I will explain the development process later, this is just a checklist of what we used, with links to each:

1. Chemicals
   a) **Kodak TMAX Professional Developer**
   b) **Kodak "Kodafix" Fixer**
   c) **Kodak Photo-Flo**
   d) **Isopropyl Alcohol** purchased from the MRL storeroom, used to clean the containers after use.
   e) **Distilled Water** from MRL storeroom.
2. Tools & Containers
   a) **Paterson 35mm Developing Kit**
   b) **Beaker** from MRL storeroom.
   c) **Kimwipes** from MRL storeroom.
   d) **Funnels** from MRL storeroom to safely pour into waste containers.
3. Extras
   a) **Film weights** used to hang film evenly when drying.
   b) **Canned air** to dry film quickly when necessary.

Figure 6: **Our developing materials.**

You now have all the materials used in our holography process.

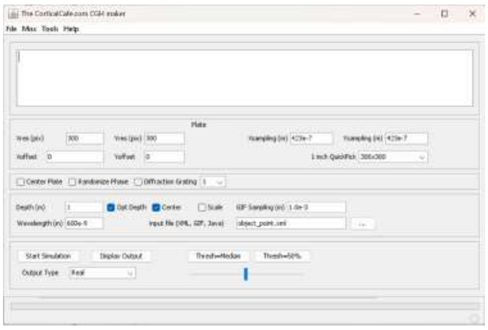
3 Designing Holograms

Creating holograms took the most time and effort throughout the project. We tried many different programs to create holograms, detailed below:

3.1 Cortical Cafe CGH (requires Java)

This code was first used by the undergrad that started this project. Here is its website. It is intended for simple laser pointers. Its interface looks like this:

Figure 7: **CorticalCafe interface.**

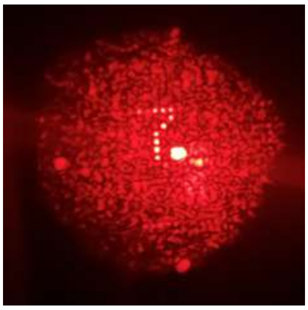


The website has instructions on how to operate the code *here*. The code takes a very long time to computer the hologram of any image larger than 30x30. We opted not to use this software because of the poor image quality and long run times.

3.1.1 Results on Film

The output (also described on the website), looks like this through a laser pointer:

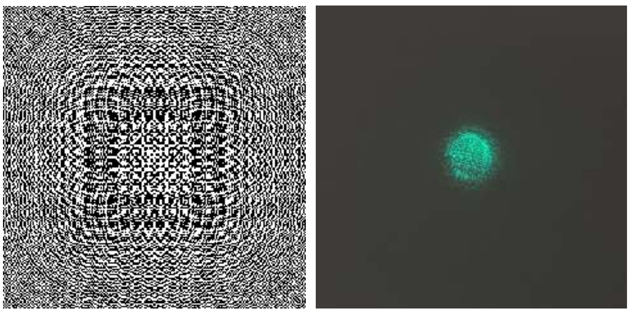
Figure 8: **Holograms from the CorticalCafe Program through a laser pointer.**



You can see this hologram has very poor quality. The input image was a pixelated number 7, which you can see on the top left and bottom right corners of the beam (bottom right is less focused).

Using a collimated beam, the hologram and output image looks like this:

Figure 9: **A CorticalCafe hologram with a collimated beam.**



We could not create any focused hologram with a collimated beam and this software.

3.2 Holoeye Pattern Generator

The Pattern Generator software can compute its own high-resolution holograms using a Fourier Transform algorithm. The program is easy to use and lets you edit the holograms through their settings. Here is the layout of the program and some tips:

Figure 10: **The SLM Pattern Generator.**

This is where you can perform many different functions, described by the Holoeye manual and here:

Elementary Optical Functions You can generate phase patterns of common optics, such as a lens, diffraction grating, and apertures. This is useful for testing your laser and its diffraction capabilities with the SLM.

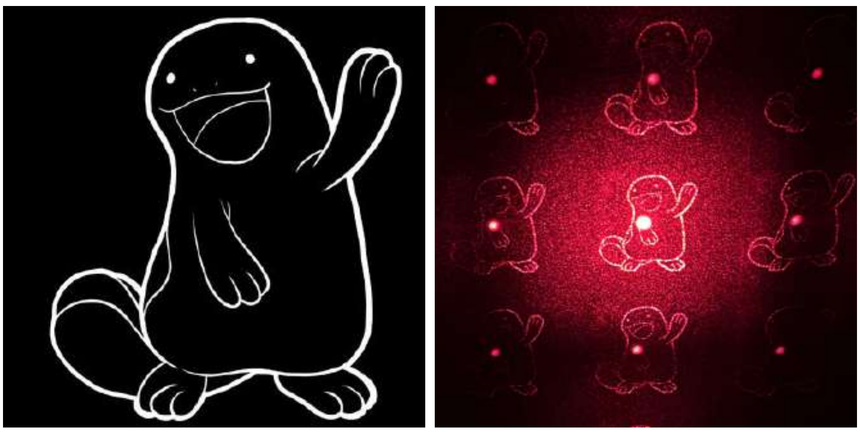
Computing Holograms First, select an image you of which would like to compute a hologram. Then, navigate to File -> Compute CGH -> select image. You will encounter a screen like this:

Figure 11: **Computing Holograms with the SLM Pattern Generator.**

The program operates on ideas of a "Unit Cell," which is how large the repeating phase pattern is. I found that small values of the slider resulted in poor hologram generation; however, many combinations of the unit cell size and slider values generated good holograms. Ensure that the image is scaled properly and any value will work.

Here is an example of computing a hologram:

Figure 12: **An example hologram and its image from the SLM.**



Note that you want the background to be black and the foreground to be white. Otherwise, you will illuminate the opposite of the image. Note the mesh-like pattern, which is caused by the geometry of the liquid crystal cells on the SLM surface.

Beam Manipulation and Lens Function Navigate to the bottom right of the interface to "Beam Manipulation." There are three functions: a vertical grating, a horizontal grating, and a fresnel lens function. Here is the overview given by Holoeye in the SLM Pattern Generator manual about each function, and the calculations used:

Figure 13: **A description of the beam manipulation menu.**

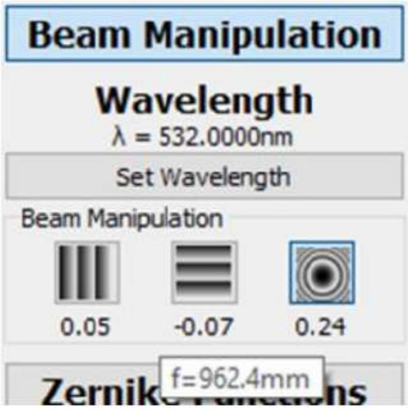


Figure 7: Beam manipulation feature.

To enable beam manipulation overlay, please click the large button 'Beam Manipulation' in the sidebar. If no incident laser light wavelength is set earlier, the software will ask for the wavelength when you apply any beam manipulation value. The incident laser light wavelength is needed to calculate the physical meaning of the unified values.

The labels below the icons show the unified value of the slider. The physical unit is shown in the hint texts of the labels.

The physical values are calculated as follows:

$$\alpha = \sin^{-1}\left[\frac{\lambda * beamSteerSliderValue}{2 * Pixelsize}\right] \quad \text{and} \quad f = \left[\frac{SLMWidthPixel * Pixelsize^2}{\lambda * beamLensSliderValue}\right]$$

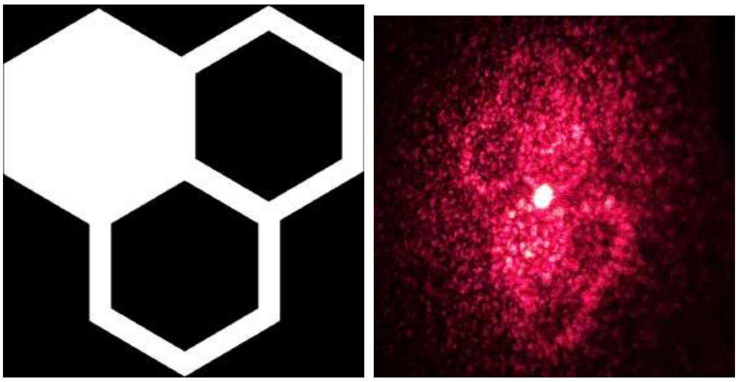
- α is the corresponding beam deviation angle between 0th and first diffraction order.
- f is the focal length of the superimposed Fresnel zone lens.
- λ is the incident laser light wavelength.
- "Pixelsize" is the physical size of the SLMs pixel, e.g. 8µm for HOLOEYE PLUTO SLMs.
- "SLMWidthPixel" is the number of pixels in the large direction of the display.
- "beamSteerSliderValue" and "beamLensSliderValue" are the unified slider values.

We frequently changed the lens parameter to create holograms that focused the beam down faster; however, values above 1, which we used, will make the image smaller. This is a parameter we needed to balance in our designs alongside others we describe in the future.

3.2.1 Results on Film

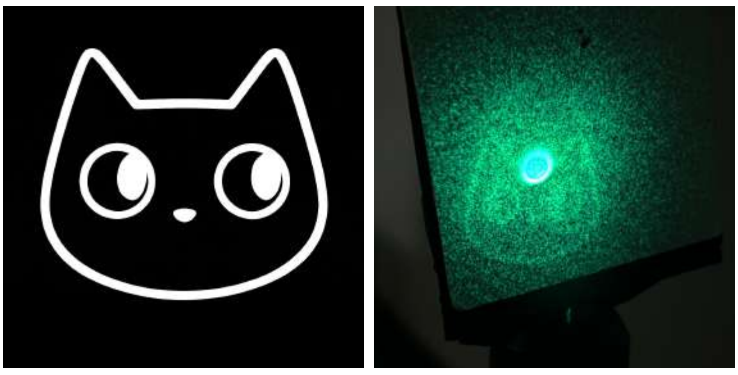
With a laser pointer, the beam diffracts like this:

Figure 14: **A hexagon pattern from SLM Pattern Generator onto film, illuminated by a laser pointer.**



With a collimated beam, the image now has one central image:

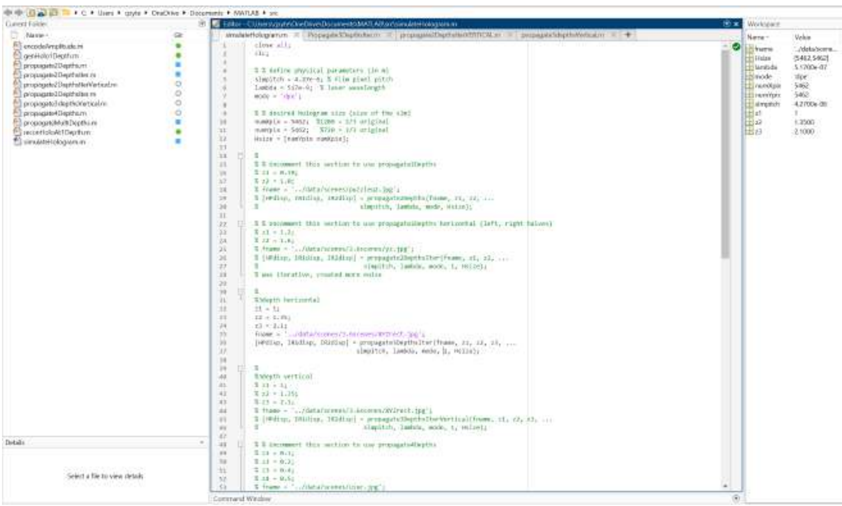
Figure 15: **The cat pattern used in LabEscape, created with the SLM Pattern Generator and a collimated beam.**



3.3 Yingsi Qin - 3D Hologram Repository in MATLAB

After creating single-depth holograms, we looked online for projects on multi-depth holography. We found work by a graduate student at Carnegie Mellon, Yingsi Qin. She created a MATLAB-based repository to solve this, using Fourier Transforms and lens functions. The interface looks like this:

Figure 16: **The MATLAB code interface.**



Enter the "pitch," or resolution, of the film printer. Ours is 4.27 μm, given by 8192 pixels wide divided by 35mm film:

$$\frac{8192\text{ pixels}}{35\text{mm}} = 234\text{ pixels/mm} \implies 4.27 * 10^{-6}\frac{\text{m}}{\text{pixel}}$$

And "lambda" is the wavelength you are using. There are many commands, here is what each of them do. The ones we created are denoted with a star (*):

1. **genHolo1Depth** - Generates a Fourier Transform of the given image and multiplies it by a lens function (concentric rings) to focus at a specific depth.
2. **propagate2Depths** - 2-depth holograms, enter both focal lengths *z1, z2*. The code will split the image in half, from left to right. The left half will be FFT'd and focused at the distance *z1*. The right half will be FFT'd and focused at the distance *z2*. Then, the two Fourier Transforms are added into one image. The image

is saved and a reconstruction is simulated. The hologram and reconstruction are then displayed in separate windows.

3. **propagate2DepthsIter** - This is an iterative version of propagate2Depths, we did not use it, as the large image computations would crash MATLAB.
4. **propagate2DepthsIterVERTICAL*** - We created this to be the same as propagate2Depths, but splits the image into top and bottom portions.
5. **propagate3DepthsIter*** - Splits the image into left, middle, and right portions, and uses FFT and lensing functions to focus. Iterative if you would like, or just set "numIter" = 1. Remember many iterations may crash MATLAB.
6. **propagate3DepthsIterVertical*** - 3 depth holograms, splitting the image into top, middle, and bottom.
7. **propagate4Depths** - 4 depths, however the normalization of many FFTs, $\frac{\text{fft1} + \text{fft2} + \text{fft3} + \text{fft4}}{4}$, leads to large background noise, i.e., poor resolution.
8. **propagateMultiDepths** - This repeatedly gave us errors during FFT calculation. We could not diagnose the problem; however, remember that many depths will create a lot of noise and work poorly onto film.
9. **reconHoloAt1Depth** - Reconstruction code to simulate the image output.
10. **SimulateHologram** - The main interface of the code, used to call each function. Uncomment each section to use each function, and make sure to pass each argument accordingly.

3.3.1 Results on Film

This program does not work with a laser pointer, only with a collimated beam. Pictures and detailed results are described in section 8.

3.4 Other Programs Tried, but not Used

We tried to use many different programs before we used the collimated beam setup. These are simply resources to check out for later use:

Computer Generated Hologram - This program was created in Mandarin, so it is sometimes difficult to trace what the code does.

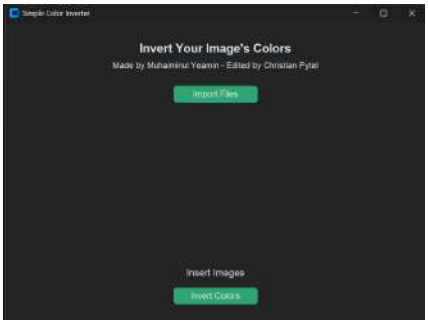
HoloPy - This program is extensive for microscopy and applications of the SLM, however it is not built directly for what we did.
PyHoloscope - This is similar to HoloPy as well.

4 Editing Images and Processing Holograms

4.1 Inverting Image Colors

Before you create your holograms, remember the images require the background to be black and the foreground to be white. I have attached an image editing notebook with a color inverter to do this, called *ImageEditorNotebook.ipynb*. The invert colors interface looks like this:

Figure 17: **Inverting image colors for hologram generation.**



After inverting the colors of the desired image, generate your hologram, then continue below.

4.2 Gamma Correction

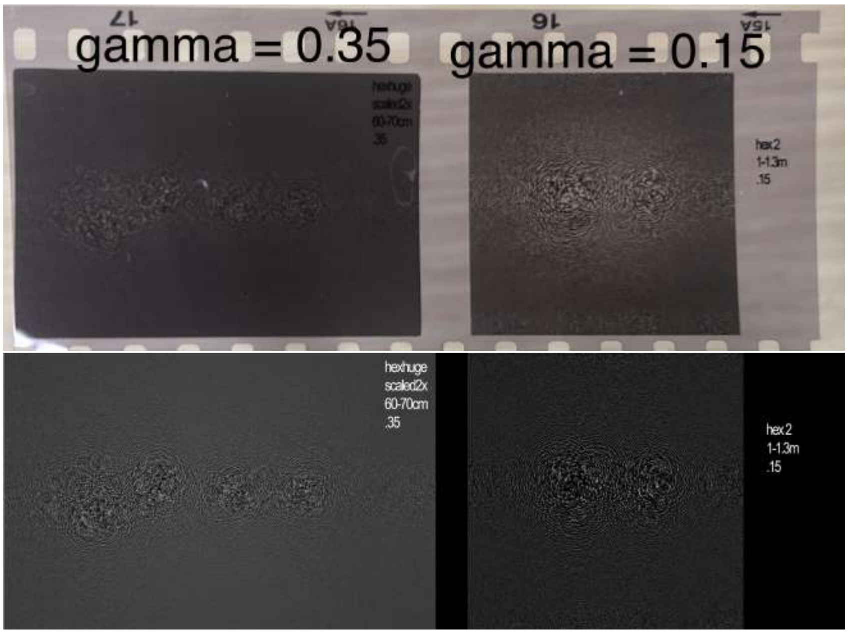
First, printing holograms onto film will be very dark. We measured $\approx 1\%$ transparency through a pure black film pattern. We rectified this by applying gamma correction, where each color value was multiplied by $x^{\frac{1}{\gamma}}$ using Python:

Figure 18: **Gamma correction with Python.**

```
def apply_gamma_correction(image_path, output_path, gamma):
    image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
    if image is None:
        print("Error: Could not load image.")
        return

    normalized = image.astype(np.float32) / 255.0

    corrected = np.power(normalized, 1.0 / gamma)

    corrected = (corrected * 255).astype(np.uint8)
    cv2.imwrite(output_path, corrected)
    print(f"Gamma corrected image saved to {output_path}")
```

Here is an example of using gamma correction to make film transparent:

Figure 19: **Using Gamma Correction on Film, with Original Images.**



4.3 Labeling in Photoshop

As UIUC students, we get the Adobe suite for free. I used Photoshop to label the images with the name of the pattern and any other characteristics. For example, the final XYZ holograms listed the focal distances z1, z2, and z3, the wavelength, the gamma value used at the top right:

Figure 20: **The XYZ hologram and its appropriate values.**

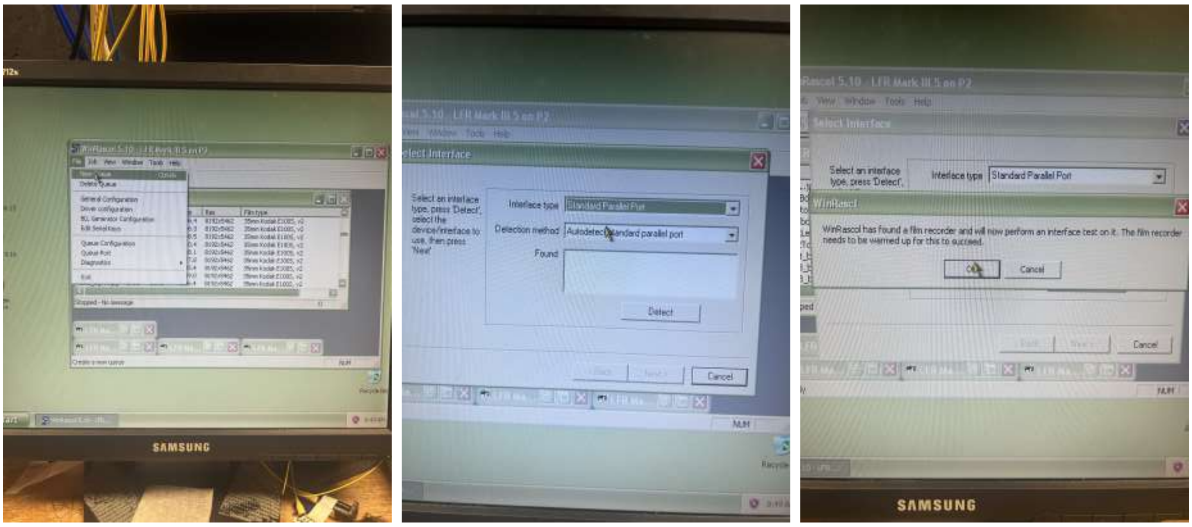
Explore different gamma values and other parameters many times with each print–you have 35 different configurations to explore each roll. Make sure the labeling is visible and that you document each one accordingly.

5 Printing Holograms

Operating the Film Printer's Software The computer we use runs Windows98. Do not connect it to the Internet; please use it ONLY for the film printer. It uses the WinRascol software on the computer.

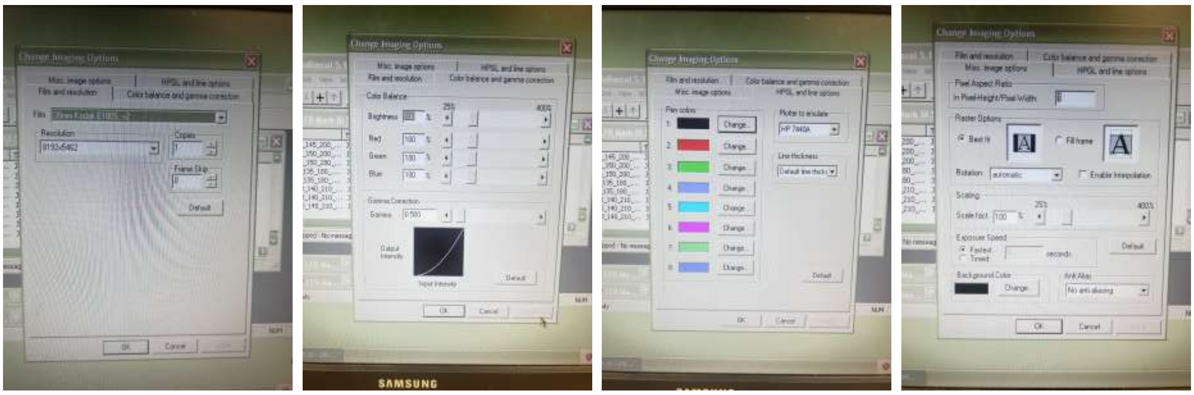
Relevant Settings To alter printings settings, you need to create a queue:

Figure 21: **Creating a Printing Queue**



Turn on the film printer and allow it to warm up. As the figure describes, create a queue using "File $\Longrightarrow$ New Queue." Then select "Standard Parallel Port." As long as the printer is warmed up, the final message will show after clicking "Detect." Then the printer will be ready to use. These are the settings we used:

Figure 22: **Settings used on the film printer.**



6 Developing Film

6.1 Materials

Alongside this document, there are videos on the Lab Server demonstrating how to develop film. Recall the list of necessary tools and chemicals for developing film in section 2.

Grab the chemicals: developer, fixer, distilled water, and PhotoFlo.
Grab your tools: film, beaker, developing tank, and the film reel.

Figure 23: **All the developing materials.**

Chemical Safety and Storage To develop film, we use diluted Kodak developer and fixer, both with harmful chemicals. Use the Selvin Lab room to develop. It has a fume hood and sink plus lab coats to wear. Use goggles if deemed necessary. Store used chemicals in safe containers, found in the MRL storeroom or online. Contact Research Safety for used chemical disposals and any other questions.

6.2 Development

With lights on Put on gloves and prepare your developing solution in a >250 mL beaker or graduated cylinder. Our TMAX Professional Developer needed to be diluted 1:4, and our tank is 250 mL, so pour 50 mL of developer into the beaker. Then pour 200 mL distilled water, and you have your developing solution.

With lights off It must be very dark in the room to ensure proper development–for example, in 388, I used a trash can and cardboard to cover the crack under the door. A good way to check if the room is dark enough is if your eyes require multiple minutes to fully adjust, and even then, the room is still dark.

Reeling the film Grab your film reel and film. Open the canister–you can use tools or simply pull the canister apart with your fingers. Pull it far enough apart that you can reach your finger into the canister and pull out the film leader, which is the beginning of the roll shown here on the right:

Figure 24: **The film leader.**

Once you have the film leader, close the canister to prevent the roll from rapidly leaving the canister. Then slide the film leader into the reel. This takes a lot of practice to do well in the dark. I have attached a video of how to do this on the lab server, or you can watch a YouTube video. Once you catch the film leader and reel in the film, you can pull the end off and throw the canister out. Place the rolled film inside the tank, and the funnel lid on top. Now take your beaker and pour in the developer. Start a timer for ≈ 6 minutes and close the lid tightly. Agitate every 30 seconds by quickly rotating the tank.

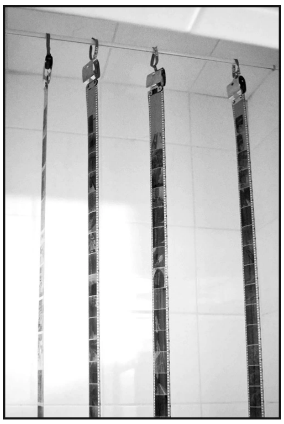
Preparing Waste Containers While the film is developing and you are agitating every 30 seconds, prepare your waste containers for the developer and fixer solution. I used an empty gallon of distilled water for each, labeled for developer or fixer.

The Fixer Solution Wash out the beaker and wipe it off. Then add 66 ml of fixer and fill the rest up to 250 ml with distilled water. Then wait until the development timer is complete.

Rinsing and Fixing Remove the from the developing tank and pour the developer into the labeled waste container, using a funnel to not spill. Then fill the tank with distilled water. Shake the container and let it soak for a minute or two, then pour the water down the sink. Finally, pour in the fixer and start a timer for ≈ 5 minutes. It is now safe to turn the lights on. Agitate every 30 seconds as well.

Final Steps After the 5 minutes, pour the fixer out into your designated waste container. Pour distilled water and a few drops of Kodak PhotoFlo into the tank. Shake the tank around to ensure it dries evenly–fixer stains are horrible! Finally, you can take your film roll to the lab and hang it vertically to dry. Tape the top of the roll to the wall or hang it like this:

Figure 25: **Vertically drying the film.**

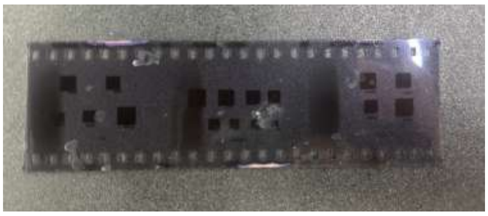


You can also directly blow off the water with canned air if you need the roll to dry quickly. The roll should dry in about an hour. Make sure to clean up the sink area of your waste containers and clean the developing tank and funnels using isopropyl alcohol and water. Your film is now successfully developed!

6.3 Troubleshooting Development Issues

Fixer Stains After developing film, you may notice white dots on each slide:

Figure 26: **White fixer stains on film.**

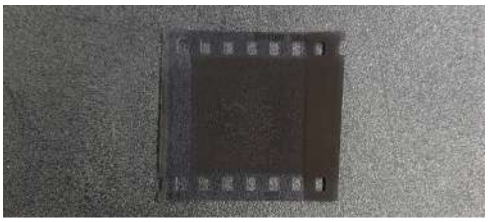


These stains indicate that drops of fixer stick to the film and wash out the affected portions. You can fix this by:

1. Ensuring your fixer is diluted 1:3 and not any stronger.
2. Forcefully agitating the fixer bath every 30 seconds or more often.
3. Rinsing multiple times, completely, with distilled water after fixing.
4. Hanging the film vertically to dry.

Dark Lines When we started to use large image files with the film printer, each slide took about 8 minutes to expose. We started to see dark lines on the left side of each slide:

Figure 27: **Dark lines on film.**



You can fix this by turning the lights off while the film printer operates or covering the front portion to minimize light exposure. Due to the age of the film printer, this is likely a mechanical issue that allows stray light to enter during exposure.

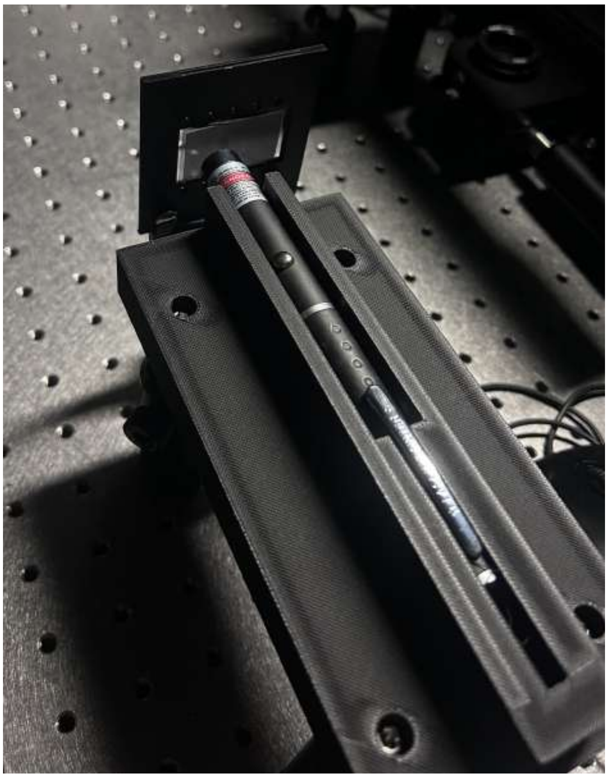
7 Testing Holograms

Once you have printed and developed your holograms, you must ensure that they work as desired.

7.1 Using a Laser Pointer

Our first way of testing was to send a laser pointer through it, like what the CorticalCafe software intends. We designed a mount for one of the laser pointers we had and attached it to the optical table:

Figure 28: **The laser pointer holder used for testing.**

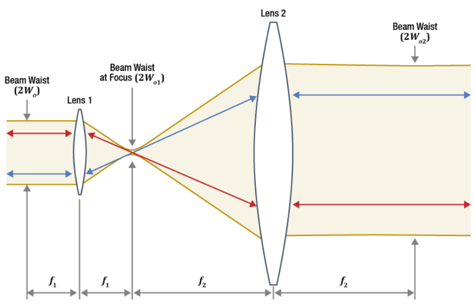


7.2 Using a Beam Expander

We found that using a laser pointer limited the size of the image we could create. If you look at the patterns above, you can see there is a ring where the picture must be contained or it will be cut off. We also wanted to have one image instead of the diagonal pairs seen with a laser pointer.

Beam Expander To expand the laser beam to one inch, we utilize a Keplerian telescope system. Here is a ray diagram:

Figure 29: **Our Beam Expander.**



We expand the beam to one inch in diameter, allowing us to illuminate a large portion of the hologram.

Alignment To align the beam expander, mount the lenses with posts and post holders, and use bases such as the Thorlabs BA1S to adjust the Z position of each lens. Place the shorter focal length lens in the beam path, and align the centers of the beam and lens. Check your alignment by using a card and letting the beam propagate. Ensure the beam is parallel to the optical axis; if not, adjust the position of the lens. Now grab the second, larger focal length lens and place it a distance $f_1 + f_2$ away. Line up the center of the second lens with the beam. With a viewing card, check if the beam moves parallel to the optical axis and maintains a constant radius. Adjust the Z position of the second lens to change focus, finishing alignment once the beam is collimated.

Refer to Section 3 for the expected images created from each software and beam orientation.

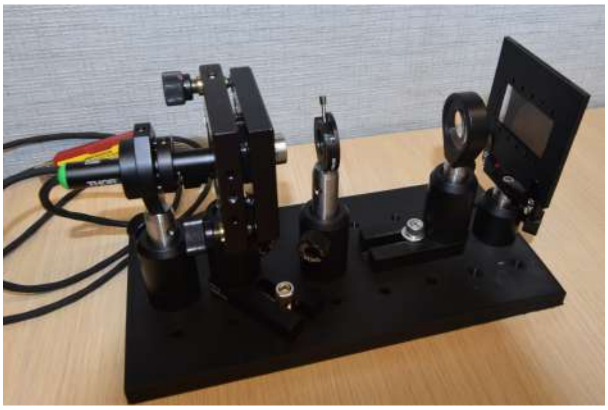
8 The XYZ Puzzle

The first puzzle we created was a three-depth image which focuses a letter X, Y, and Z at different depths. The distances where the letters appear (in centimeters) are the solutions to the puzzle. Our setup has been part of the traveling *LabEscape* room, entertaining scientists at CLEO and the APS March Meeting.

8.1 Optical Setup

Due to the travel requirement, we ensured portability through a 3D printed plastic breadboard and compact optics:

Figure 30: **The Optical Setup used in LabEscape.**



Laser The laser is a Thorlabs PL251 USB laser. Its spectrum is centered around 517 nm and employs a typical power of 4.5 mW and a spot size of about 3mm. We mount it using a post and post holder, a FMP1 optic mount and a KAD11NT Kinematic adapter. Here is a close-up of the laser mount from Thorlabs:

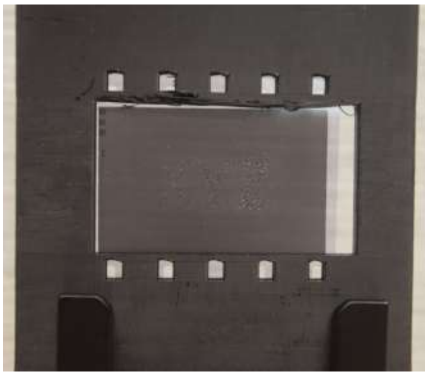
Figure 31: **The laser mount.**

Operation and Safety The laser is switched on and off through the USB connector. A green light will indicate that the laser is on. The Thorlabs PL251 is a Class 3R laser, indicating the beam can cause harm if viewed directly. While the beam is on, never lift the breadboard or move the film, as the beam can reflect off surfaces and enter the eye. Do not look into the beam path to check alignment, always use a viewing card to do so.

Beam Expander The beam is expanded to one inch in diameter by our Keplerian telescope system, consisting of a Thorlabs C230TMD-A collimation package ($f = 4.5$ mm) and a Thorlabs LA1509-A lens ($f = 100$ mm). This corresponds to a magnification of $M = \frac{100}{4.5} = 22.2$ times, meaning we expand the beam to approximately 2 inches. We used the collimation package due to a shortage of fast lenses. The outer portions of the beam that are not collimated by the LA1509-A lens diverge as stray light.

Hologram The last optic is the hologram printed on film. The film is held with a 3D-printed mount, and fastened to the breadboard using a Thorlabs FH2D filter holder. The film holder design is in the Lab Server under "Christian/LabEscapeHolography/-CAD/LabEscape Optics/FilmHolder.stl" and was designed on Fusion360. It is printed with PLA and holds the film like this:

Figure 32: **The film holder and XYZ hologram.**



Optical Breadboard For portability, we designed a 3D printed optical breadboard instead of a Thorlabs aluminum board. The breadboard measures 8.5" x 4" and has 24 holes, each spaced one inch apart in an 8 x 3 grid. We used the Physics Machine Shop to tap the holes into $\frac{1}{4}$-20 threads for posts and post holders.

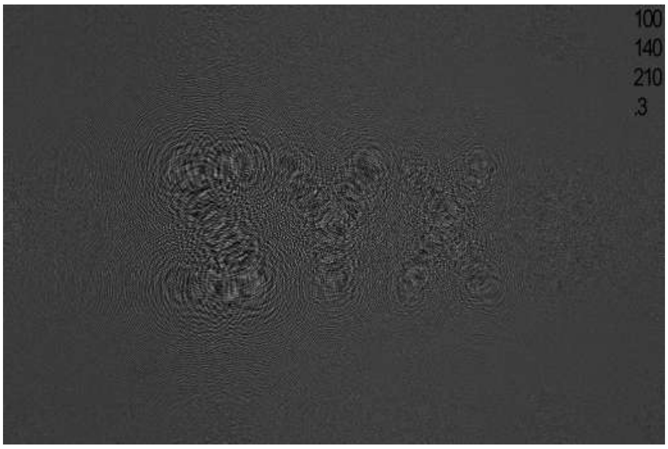
8.2 Hologram Design

Our final holograms used in LabEscape utilized the 3DHologram repository and these settings in MATLAB:

Parameter	Value	Description
slmpitch:	$4.27 * 10^{-6}$	SLM Resolution
lambda:	$517 * 10^{-9}$	Wavelength
z1:	1.0	1st Focusing Distance
z2:	1.4	2nd Focusing Distance
z3:	2.1	3rd Focusing Distance
Gamma:	0.3	Gamma Correction Value

We indicated these settings using Photoshop, making a label at the top right:

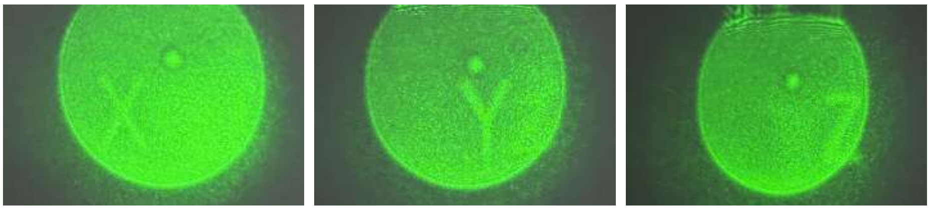
Figure 33: **Our final holograms used in LabEscape.**



8.3 Final Results

Here are the results of the hologram, capturing the beam on a viewing card at specific distances after the film:

Figure 34: **Final Results as seen in *LabEscape*.**

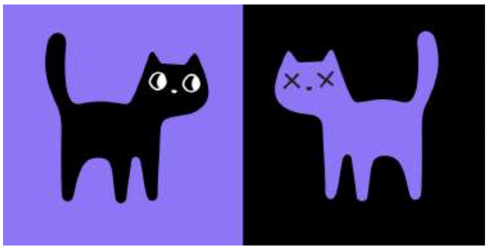


Focusing Distances We originally created this hologram to focus at 100 cm, 140 cm, and 210 cm; however, the realized focusing distances turned out to be 50 cm, 75 cm, and 120 cm. We believe this is due to the gamma correction we impose, thus changing the amplitude grating. The distances 100, 140, and 210 cm with $\gamma = .3$ were found to create the best image while maintaining a suitable path length for the room.

9 "9 Lives" Cat Puzzle

Created in October 2025, we designed this puzzle to revamp a previous one. The puzzle involves using a laser pointer, cat hologram, and diffraction gratings to create five images of cats, therefore creating "45 lives." The previous puzzle used a laser pointer and hologram created by Optica to replicate their cat logo:

Figure 35: **The Optica cat puzzle.**



We changed the puzzle to a single-depth image of just the cat head. We tried to use the 3D Hologram program to generate a single-depth image, but had repeated issues, so we used the SLM Pattern Generator program instead.

9.1 Optical Setup

This puzzle required more optics than the XYZ puzzle due to spatial filtering and beam blocks:

Figure 36: **The 9 Lives Optical Setup.**

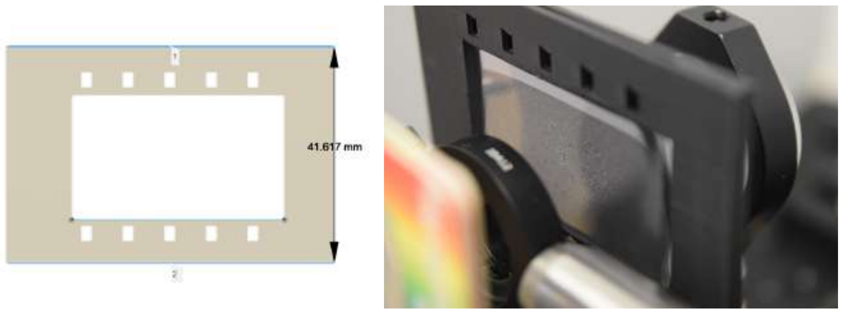
Laser We used the same Thorlabs PL251 USB laser as in the XYZ puzzle.

Iris We use an iris to limit the beam size as it enters the beam expander. This maintains the degree of freedom to expand the beam and creates a circular wavefront.

Beam Expander We use a Keplerian telescope systems with one $f = 12.5$ mm lens and a $f = 50$ mm lens. This corresponds to a magnification of $M = \frac{50}{12.5} = 4$ times expansion.

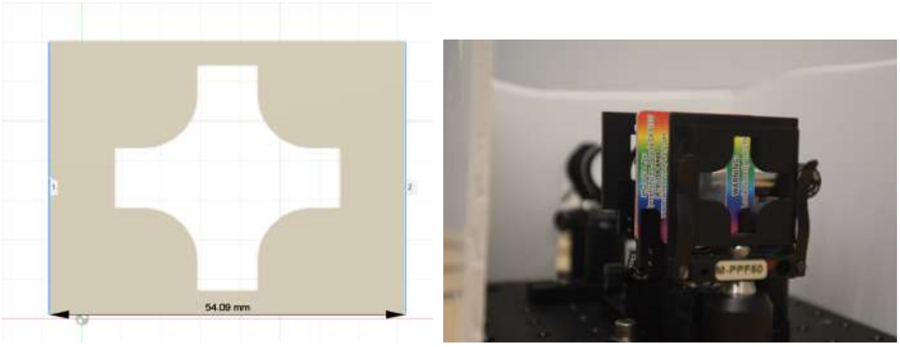
Hologram Mount To conserve height, we utilized a shorter 3D-printed film mount:

Figure 37: **The shorter film holder used in the 9 Lives puzzle.**



Diffraction Grating & Beam Block We use two 500 lines/mm diffraction gratings, aligned perpendicular to each other, to split the beam into a mesh of 9 cats. Due to the poor image brightness of the diagonal elements, we block the cats appearing at the corners to create 5 propagating images. We designed a 3D printed beam block and implemented it as shown:

Figure 38: **The beam block and grating used in the 9 Lives puzzle.**

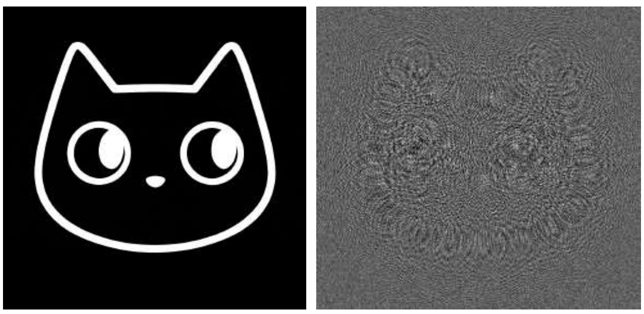


9.2 Hologram Design

Our goal was to create a single-depth cat image that did not create the images split across the diagonal; instead, we wanted one image that propagates with the beam. We first opted to use the 3DHologram repository and write a function to generate single-depth holograms. It is called "propagate1Depth" and employs the same machinery as the multi-depth functions. We obtained poor results with these holograms and opted not to use them, instead using the SLM Pattern Generator.

Using SLM Pattern Generator Downloading and use of SLM Pattern Generator is described earlier in section 2. Here is the cat image and hologram we used:

Figure 39: **The Optica cat outline.**

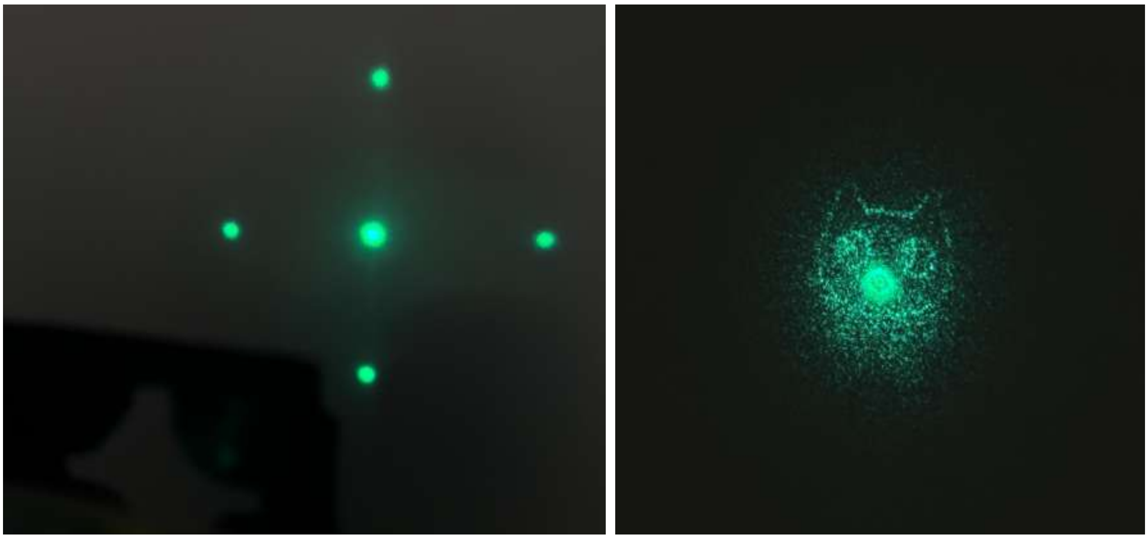


We iterated over changing the lens function value in the "Beam Manipulation" tab and the gamma correction value used. This balances the image size and focusing distance with the contrast and efficiency of the hologram. We found that lens values around 1.4-1.8 and γ values of 0.125 gave the best results.

9.3 Final Results

Participants use a white viewing card or paper to view the central beam and the 4 cardinal directions diffracted by the grating. As they move along with the beam, the cats begin to focus at around 50 cm. Since there are 5 cats, each with "9 Lives," the answer to the puzzle is 45. Figure 40 demonstrates the beam a few centimeters after the grating and the focused cat image.

Figure 40: **The beam after the grating and focused cat image.**



10 Appendix A: Holographic Plates

If you would like to explore using holographic plates, those in the past used PFG-01 plates from Integraf and the necessary chemicals, which are contained in the JD-2 kit *link*.

We purchased the U08M plates from Ultimate Holography, along with the Ultimate Developer and Bleach. I recommend buying the Integraf plates if possible, as they caused an issue with UIUC Purchasing.

We contacted Julio and attempted to recreate his setup. He describes his work in his thesis *here*. He exposed his holographic plates as shown:

Figure 41: **Exposing Holographic Plates**